

Vaccination and child mortality in 2020

Higher vaccination rates generally coincide with lower child mortality across the 191 countries with both indicators. The relationship is moderately negative (Pearson $r = -0.52$), with substantial variation among countries at similar vaccination rates.

What the indicators measure

Vaccination rate (%): the share of one-year-olds who have received at least one of the following vaccinations, as defined by the report author: BCG, DTP3, HepB3, Hib3, measles 1st, measles 2nd, PCV3, Pol3, RotaC. This is not the share completing every listed vaccination or a full schedule.

Child mortality: deaths before age five per 1,000 live births, using the supplied dataset. For example, 59.66 per 1,000 is equivalent to about 5.97%.

The author-provided vaccination definition supplements the provisional metadata notes printed in the original figures.

How the analysis was conducted

Python selected the **2020** column in each CSV and joined records using the **geo** country code. No values were interpolated or imputed. All 191 vaccination observations matched mortality observations; three additional mortality observations were excluded from the paired analysis.

The scatterplot gives each country equal weight. Pearson correlation measures the linear association; Spearman rank correlation is also negative (-0.44). These are descriptive comparisons, not estimates of a vaccine effect.

The maps use Natural Earth 1:110m outlines. Some small countries are absent at this scale. Grey areas have no matched value; outlines are contemporary rather than historical 2020 boundaries.

The three highlighted countries

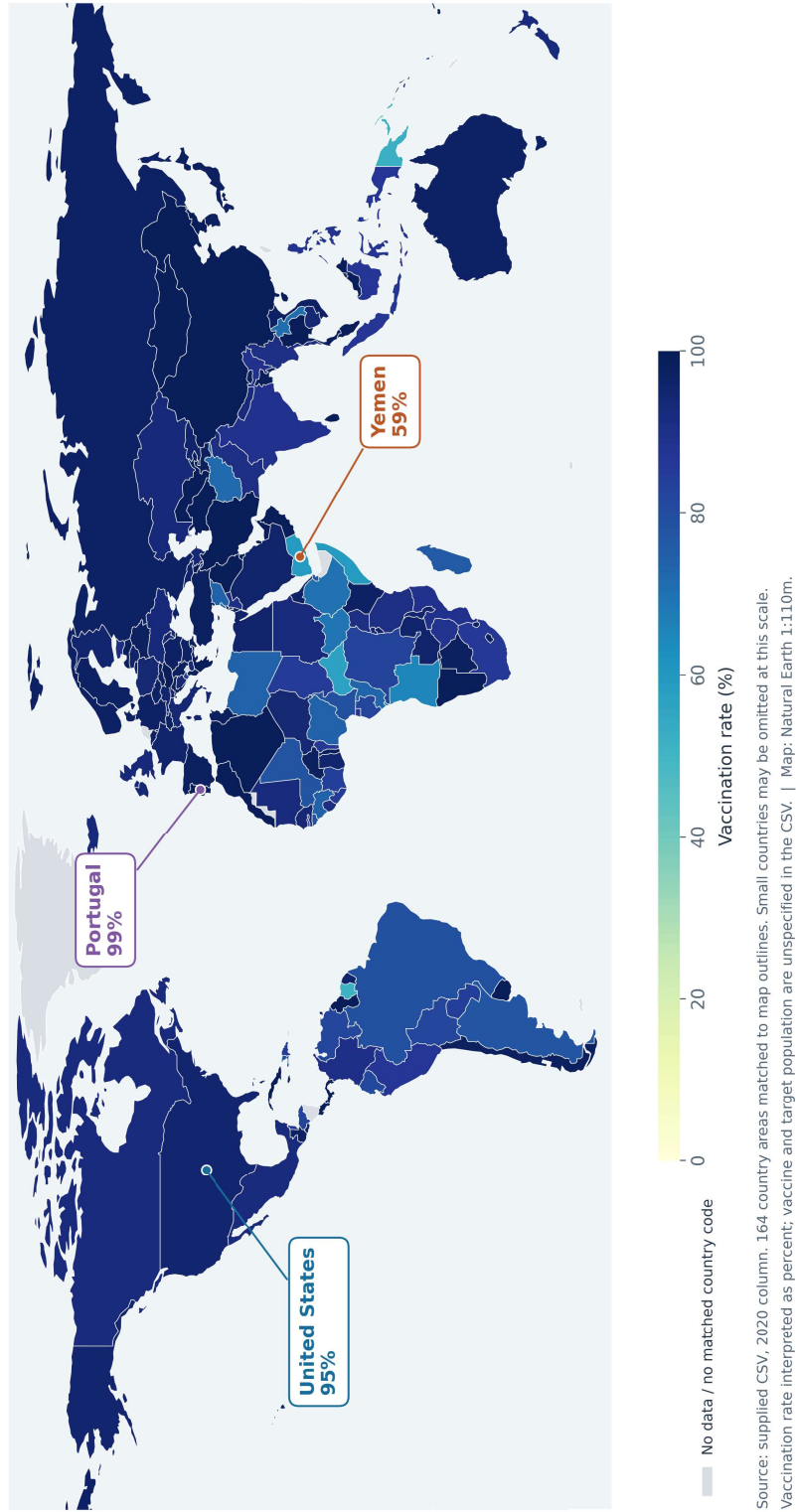
Country	Vaccination rate	Child mortality / 1,000 births
United States	95%	6.36
Portugal	99%	2.89
Yemen	59%	59.66

FIGURE 1 | GEOGRAPHIC DISTRIBUTION

Vaccination rates across the world

Vaccination rate | 2020

Vaccination rate (%)



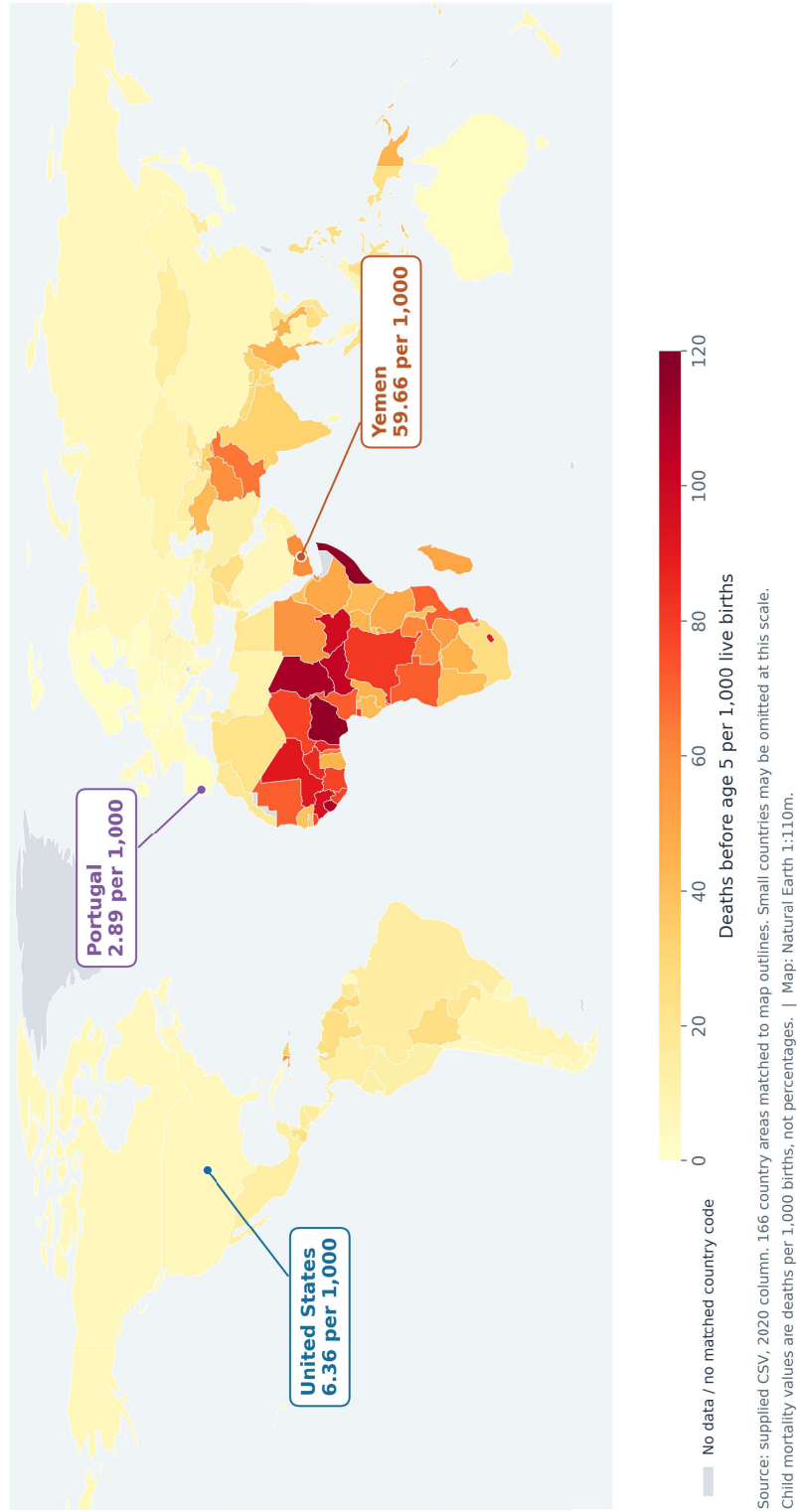
Interpretation. Vaccination rates are high in many countries, while lower values are visible in parts of Africa and other regions. Across the 191 observations, the median is **96%** and the range is **51%-99%**. Portugal (99%) and the United States (95%) lie near the upper end; Yemen (59%) is well below the median. These values describe receipt of at least one listed vaccination, so a high value does not demonstrate completion of the full schedule.

FIGURE 2 | GEOGRAPHIC DISTRIBUTION

Child mortality across the world

Child mortality under age 5 | 2020

Deaths before age 5 per 1,000 live births



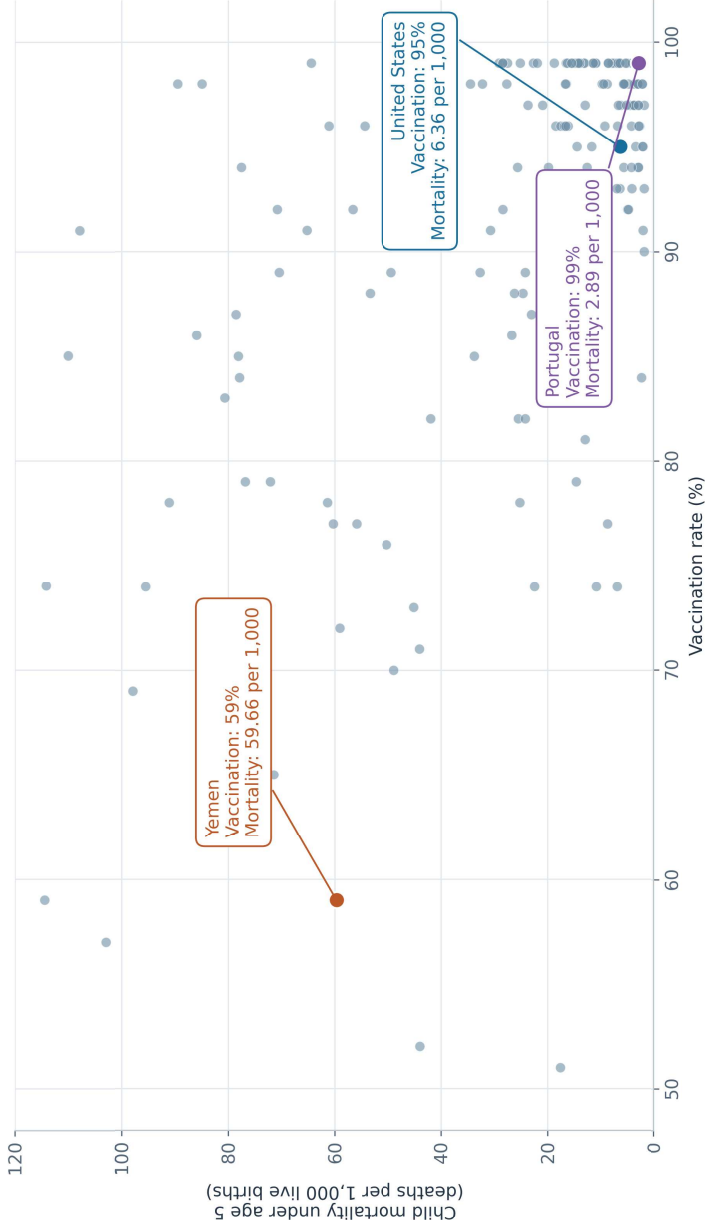
Interpretation. Higher child mortality is visible across many countries in sub-Saharan Africa and selected countries elsewhere. Among the 191 matched countries, the median is **15.50 deaths per 1,000 births**, with a range of **1.85-114.39**. Portugal (2.89) and the United States (6.36) are below that median; Yemen (59.66) is above it. Yemen's rate is about 9.4 times the United States rate and 20.6 times Portugal's rate. These are comparisons of observed rates.

FIGURE 3 | RELATIONSHIP AND INTERPRETATION

A negative association, with substantial variation

Vaccination and child mortality | 2020

191 countries with both indicators • each point represents one country



Source: supplied CSV files, matched by country code using 2020 values. Missing pairs excluded; no imputation.
Vaccination rate interpreted as percent; vaccine and target population are unspecified in the CSV.
This cross-country association does not establish causation.

What the pattern shows

The overall downward pattern is consistent with higher vaccination rates accompanying lower child mortality. Pearson $r = -0.52$ and Spearman correlation $= -0.44$ summarize this association.

What the highlighted countries show

Portugal and the United States combine high vaccination rates with low mortality. Yemen combines a lower vaccination rate with higher mortality. These three examples fit the overall pattern, but do not explain every country.

What cannot be concluded

This is a single-year, country-level comparison, not a causal study. It does not align the same birth cohort across the indicators or adjust for healthcare access, income, sanitation, nutrition, conflict, or data quality. Such factors could influence both indicators. Country averages also conceal variation within countries.

Data provenance: vacc_rate.csv and child_mortality_0_5_year-olds_dying_per_1000_born.csv, 2020 columns; vaccination definition supplied by the report author. Underlying source, estimation methods and uncertainty intervals were not supplied. Figures were generated in Python using pandas and matplotlib; map boundaries: Natural Earth v5.1.2 (public domain).

Impressed by AI's capability for simple analysis

Summary

This analysis explored the relationship between vaccination rates and child mortality across 191 countries in a dataset provided by Gapminder. The United States, Portugal and Yemen were selected as representative countries within this dataset. There is a moderate negative correlation between vaccination rate and childhood mortality rate. Causation was not assessed and should not be assumed. The data reflect high variability among counties with similar vaccination rates and many variables impacting both vaccination rates and childhood mortality were not included in the analysis.

Working with AI

I utilized ChatGPT (paid version) to produce the first 3 pages of the report above. The three figures produced were well done and required very little manipulation. I only requested a single change to make one label easier to see. I like that Chat automatically chose a different color scheme for the two maps to keep the report engaging for the viewer. I asked Chat to include general descriptions and interpretations of the data and analysis and was generally impressed by the content the model was able to provide, particularly around limitations of the data and how to prudently interpret the analysis. Chat also appears to have used appropriate statistical methods to analyze the data without any specific prompts on which statistics to use. I was pleasantly surprised by this and wonder how well the model can continue to do so with datasets of increasing complexity.

I also tried Gemini (free version) using identical data and prompts. I liked the visual layout of the report from Gemini. However, when manually spot checking the data, it's clear that the free version of Gemini's report contained inaccurate information. I was surprised that Gemini wasn't able to pull accurate values directly from a very clean and complete dataset. This model also did not provide any context on limitations and I feel it overconfidently stated certain relationships in the data. I suspect the paid version would be better but this exercise makes me curious about what makes different models produce different results when completing simple tasks from clean datasets.